SANSKRUTI NAIK

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PROFESSIONAL SUMMARY

B.Tech (AI & ML) student graduating October 2026, building end-to-end AI and mobile products, from LLM/RAG pipelines and CNN inference to real-time encrypted Android/iOS apps. Built an E2EE messaging platform and a 92%-accuracy CNN classifier with 30% faster inference. GDG Core Member and national hackathon lead — seeking an engineering role building user-facing AI products.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, R, SQL
AI / ML: LangChain, Hugging Face, OpenAI API, TensorFlow, PyTorch, Scikit-learn, RAG, LLMs, CNNs, Grad-CAM, SHAP
Frontend / Mobile: React.js, Flutter, Android, Figma, Blender, Canva
Security & Real-Time Systems: OMEMO2 / Signal Protocol (E2EE), WebRTC, XMPP, WebCrypto
Deployment & Tools: FastAPI, REST APIs, Docker, Git, Google Colab, Vercel, PostGIS, Nginx, Node-RED
Core Concepts: Multimodal AI, XAI, Satellite Imagery Analysis, Object Detection, NLP, DSA, Product Management for AI

EXPERIENCE

AI / ML Intern · Endureva Innovations LLP *Jan 2026 – June 2026*

- Built and deployed a CNN-based plant disease classifier (92% accuracy, 50K images, 38 classes) served via FastAPI with <200ms inference latency.
- Cut model inference time 30% through quantization and batch optimization; owned the ML pipeline end-to-end from preprocessing to production.
- Built an offline-first Restaurant POS app (Flutter + FastAPI + SQLite) with JWT role-based access and a live analytics dashboard.
- Presented model results and technical trade-offs to non-technical stakeholders, translating outputs into business decisions.

PROJECTS

SecureChat — End-to-End Encrypted Messaging Platform *2026 – Present*

React 18 · XMPP/ejabberd · OMEMO2 (Signal Protocol) · WebRTC · PostgreSQL · Docker · Capacitor

- Architected an E2EE messaging platform with OMEMO2 (Signal Protocol) Double Ratchet encryption across messages, media, and calls.
- Built a local-first storage layer (IndexedDB + encrypted key store) for offline message history and media caching, so the app is fully usable without persistent connectivity.
- Implemented WebRTC voice/video calling (coturn TURN/STUN) and encrypted file sharing up to 500MB via chunked AES-GCM streaming.
- Shipped Android/iOS builds from a single React codebase (Capacitor) and containerized the full stack (ejabberd, PostgreSQL, Nginx, coturn) with Docker Compose.

RAG AI Model — Civil Tender Document Intelligence *2026 – Present*

Python · LangChain · RAG · LLM · FastAPI · Vector Embeddings

- Designed the RAG retrieval pipeline and LLM prompting logic that answers eligibility, EMD, scope, risk, and penalty queries with clause-level citations, cutting tender analysis from hours to minutes.
- Built the risk-finder and go/no-go decision engine on top of the retrieval layer for end-to-end procurement intelligence.

Imagery-Based Unauthorised Construction Detection System *2026 – Present*

Python · YOLOv8 · Sentinel-1/2 · SAR · PostGIS · NDVI/NDBI

- Architected the PostGIS spatial database and Web GIS dashboard for a 3-model ensemble (Sentinel-2, SAR, Cartosat-3 fusion) detecting pipeline encroachments, targeting ~95% accuracy and a <4% false positive rate.
- Designed the 5-day automated satellite monitoring cycle with SMS/email alerting.

XAI-Care: Explainable Multimodal AI for Elderly Assistance *2025 – 2026*

Python · TensorFlow · FastAPI · Grad-CAM · SHAP · Flutter · Wearable Sensors

- Built the Flutter caregiver dashboard and multimodal fusion logic (wearable sensors + camera) for real-time fall detection.
- Applied Grad-CAM and SHAP to surface explainable, actionable alerts for non-technical caregivers.

Water Survey — Flutter Field Application *2026*

Flutter · SQLite · REST API · Offline-First

- Built the mobile app enabling field surveyors to capture water connection data offline with geo-tagging and structured export.
- Implemented auto-sync on reconnect and an offline-first data architecture for unreliable field network conditions.

AI Plant Disease Classifier *2026*

Python · TensorFlow · PyTorch · OpenCV · Grad-CAM · Flutter

- Built Grad-CAM explainability into a 92%-accuracy, 38-class CNN classifier, deployed via Flutter for rural, low-connectivity use.

Restaurant POS System (Offline-First) *2026*

Flutter · FastAPI · SQLite · JWT Authentication

- Built the Flutter frontend and JWT-based role authentication (Admin/Staff) for a fully offline POS system.
- Implemented inventory management and a real-time revenue analytics dashboard.

EDUCATION

B.Tech — AI & Machine Learning *October 2026*
Sanjay Ghodawat University · CGPA: 8.36/10

12th Standard — Science *2022*
Vivekanand College, Kolhapur

ACHIEVEMENTS, CERTIFICATIONS & LANGUAGES

- 3rd Prize — Smart India Hackathon (College Level): Sign language learning app for English & Gujarati Sign Language conversion.
- Management Team Lead — Prabal Hackathon (GDG SGU): Led a 40-hour national hackathon with 600+ participants.
- Core Member — Google Developer Group SGU: Facilitated AI/ML workshops, design sprints, and peer learning sessions.
- Certifications: ISRO Remote Sensing & Digital Image Analysis · Software Engineering Simulation (Accenture) · C++ OOP
- Languages: English (Fluent) · Hindi (Fluent) · Marathi (Fluent) · German (Basic)